



PRODUCT REQUIREMENTS DOCUMENT

AI Boardroom

Enterprise Multi-Agent Decision Intelligence Platform

A virtual executive boardroom where specialized AI agents collaborate, debate, and deliver board-grade analysis on any idea, document, or dataset.

PLATFORM	AI Boardroom
VERSION	1.0 — Initial Release
CATEGORY	Enterprise AI · Multi-Agent Systems · Decision Intelligence
STACK	Next.js · FastAPI · LangGraph · OpenAI · Anthropic · Qdrant · PostgreSQL
AUDIENCE	Founders · Investors · Product Managers · Analysts · Enterprises
STATUS	Pre-Build — Architecture Approved
DOCUMENT	Product Requirements Document (PRD)

"The goal is NOT to build another chatbot. The goal is to create an AI-powered boardroom where users receive expert-level, multi-perspective analysis — structured, explainable, and actionable."



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01 Executive Summary

AI Boardroom is a production-grade, collaborative multi-agent decision intelligence platform. Users submit ideas, business proposals, research papers, datasets, marketing strategies, or any project-related content. Instead of a single AI reply, the platform assembles a virtual executive board of specialized AI agents that analyze the submission from distinct professional perspectives — simultaneously, transparently, and accountably.

The platform simulates how real executive teams evaluate opportunities, risks, products, and investments — producing individual expert reports, a structured boardroom discussion, and a final executive recommendation complete with confidence scores and an action plan.

9	3	10+	1M+
AI Agents	Round Model	Report Types	Scale Target
Specialized experts	Max collaboration rounds	Per boardroom session	Concurrent users

PLATFORM PILLAR	DESCRIPTION
Multi-Agent Collaboration	9 specialized agents working in parallel with shared memory
Dynamic Orchestration	CEO Agent selects only relevant agents per input type
RAG-Powered Memory	Documents chunked, embedded, retrieved — never reprocessed
Three-Round Discussion	Independent → Peer Review → Challenge & Refine → Consensus
Board-Grade Reporting	PDF, DOCX, PPT output styled as executive deliverables
Immersive 3D Interface	React Three Fiber boardroom with live agent status visualization
Multi-Provider AI	OpenAI + Anthropic + Gemini with intelligent routing & fallback
Enterprise Scalability	Kubernetes-orchestrated, designed for 1M+ users without redesign

★ AI Boardroom is not a chatbot. It is a Virtual Executive Board.

02 Problem Statement

2.1 The Challenge

PROBLEM	DESCRIPTION
Single-Perspective AI Outputs	All mainstream AI tools return one response from one model — no depth, no debate, no opposing views.
No Expert Collaboration	Real decisions need strategic, financial, technical, customer, and risk analysis simultaneously. This requires teams of experts — expensive and slow.

PROBLEM	DESCRIPTION
Opaque AI Reasoning	Existing tools provide no transparency into how a recommendation was formed. Decision-makers cannot audit or trust the output.
Generic, Non-Deliverable Outputs	AI chat responses are ephemeral. They cannot be handed to a board, investor, or client as a professional deliverable.
No Structured Multi-Round Analysis	There is no AI system that replicates the back-and-forth of a real boardroom — where experts challenge each other and refine positions.
Cost of Human Expert Panels	Hiring strategy consultants, financial advisors, legal counsel, and technical reviewers for every decision is prohibitively expensive for most organizations.

2.2 Impact

These gaps leave founders making uninformed bets, analysts relying on single-model outputs, and enterprises paying enormous consulting fees for analysis that AI Boardroom can deliver in minutes. The opportunity is to democratize access to board-level intelligence.

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Goals & Success Criteria

GOAL	SUCCESS CRITERIA	STATUS
Multi-Agent Analysis	User submits input → all relevant agents activate and produce reports	✓ Core Requirement
Parallel Execution	All agents execute simultaneously — not sequentially	✓ Core Requirement
Three-Round Collaboration	Agents complete Independent → Review → Challenge rounds	✓ Core Requirement
Dynamic Agent Selection	Only relevant agents activate per input type — no unnecessary execution	✓ Core Requirement
RAG Memory System	Documents chunked, embedded in Qdrant, retrieved by agents — not reprocessed	✓ Core Requirement
Confidence Scoring	Reviewer Agent produces GO / GO WITH MODIFICATIONS / NOT RECOMMENDED	✓ Core Requirement
Report Export	PDF, DOCX, and PPT export for all report types	✓ Core Requirement
Live Boardroom View	Real-time 3D visualization of agent status during analysis	✓ Core Requirement
Authentication	Google OAuth + Microsoft OAuth + email/password with RBAC	✓ Core Requirement
Audit Trails	Full prompt logs, decision logs, session history — permanently stored	✓ Core Requirement
API Response Time	< 200 ms per endpoint under normal load	✓ NFR Target
Scalability	Architecture supports 1M+ users without redesign	✓ NFR Target

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Target Users

4.1 Primary Users

USER TYPE	SEGMENT	PRIMARY USE CASE
Entrepreneurs & Founders	Individuals	Validate startup ideas, business models, GTM strategy, investor pitch readiness
Investors & Analysts	Individuals	Due diligence, deal evaluation, competitive landscape, financial projection review
Product Managers	Organizations	Product concept validation, feasibility, customer fit, technical scoping
Researchers & Academics	Individuals	Research paper analysis, dataset exploration, literature review synthesis
Strategy Consultants	Organizations	Multi-perspective client reports, rapid analysis generation, scenario modelling

4.2 Secondary Users

USER TYPE	USE CASE
Enterprises & Boards	Strategic planning, operational risk reviews, M&A; analysis, organizational decisions
Startups & SMBs	Affordable access to board-level intelligence without hiring consulting firms
Educational Institutions	Teaching decision frameworks, business analysis, research methodology
Government & Public Sector	Policy impact analysis, infrastructure planning, risk assessments

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User Flow

The complete end-to-end journey from first visit to final executive report.

1

Authentication

User arrives at the premium login page. Sign up or log in via email/password, Google OAuth, or Microsoft OAuth. Session is created securely with role assignment (Admin / User).

2

Dashboard

Post-login dashboard displays Recent Projects, Saved Sessions, Analytics overview, and the primary CTA: Create New Analysis. Communicates professionalism and intelligence immediately.

3

Submit Input

User describes their idea via free-text or uploads files (PDF, DOCX, PPT, CSV, XLSX, JSON, TXT). Mixed inputs are supported — combine a business plan PDF with a financial CSV and a text description.

4

Preprocessing

The backend extracts text, tables, and metadata from all uploaded files. Content is chunked, embedded via OpenAI embeddings, and stored in Qdrant for agent retrieval.

5

CEO Orchestration

The CEO Agent (LangGraph) analyses the input, classifies the problem type, selects the relevant subset of agents, creates a shared workspace, and initiates the boardroom session.

6

Round 1 — Independent Analysis

All selected agents execute in parallel. Each produces an independent expert report from its unique perspective without seeing other agents' findings.

7

Round 2 — Peer Review

Every agent reads the other agents' reports and updates its own analysis based on new information, identifying agreements and initial disagreements.

8

Round 3 — Challenge & Refine

Agents challenge contradictions, reinforce consensus positions, and raise final concerns. The Devil's Advocate Agent tests assumptions and surfaces failure scenarios.

9

Reviewer Synthesis

The Reviewer Agent collects all outputs, resolves conflicts, builds consensus, assigns confidence scores to each claim, and formulates the final recommendation.

10

Report Generation & Delivery

Individual expert reports + boardroom discussion log + executive summary are generated in PDF, DOCX, and PPT formats. The user views and downloads all deliverables from the dashboard.

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Input Types

INPUT TYPE	CATEGORY	EXAMPLES & USE CASES
Free-Text Idea	Text	Plain-language description of any concept, idea, or business proposal
PDF Documents	Document	Business plans, research papers, strategies, whitepapers, legal docs
DOCX Files	Document	Word documents, memos, reports, project proposals
PPT Presentations	Document	Slide decks, investor pitches, strategy presentations
CSV / Excel Datasets	Data	Financial models, market data, survey results, operational metrics
JSON Exports	Data	API exports, database dumps, structured application data
TXT Files	Text	Raw text, code documentation, unstructured notes
Mixed Input	Combined	Any combination of the above in a single boardroom session

■ Max file size: 50 MB per file · 200 MB per session · Preprocessing is automatic.

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Agent Architecture

AI Boardroom operates through nine purpose-built agents. The CEO Agent orchestrates the workflow but never makes the final call. Expert agents analyze independently, then collaborate through a three-round discussion. The Reviewer Agent synthesizes consensus and issues the recommendation.

★ CEO Agent › Classify input type › Select agent team › Init shared workspace › Manage workflow	▲ Strategic Agent › Vision & direction › Long-term potential › Strategic opportunities › Market positioning	◆ Business Agent › Market analysis › Competitor landscape › Business model review › Revenue opportunities
■ Financial Agent › Cost analysis › ROI & projections › Profitability assessment › Budget planning	■ Technical Agent › Feasibility analysis › Architecture review › Technology selection › Scalability planning	■ Customer Agent › User persona mapping › Adoption modelling › Customer needs analysis › UX recommendations
■ Risk Agent › Legal risk mapping › Operational risks › Financial risk scoring › Compliance checks	■ Data Agent › Dataset profiling › Trend detection › Statistical forecasting › Data quality audit	✗ Devil's Advocate › Challenge assumptions › Failure scenario sim › Counter-arguments › Weakness detection



Reviewer Agent

- > Collect all findings
- > Detect conflicts
- > Build consensus
- > Confidence scoring

AGENT	ROLE & RESPONSIBILITIES
CEO Agent	Orchestrator — never the final decision-maker. Understands context, classifies problem, selects agents, manages shared workspace, monitors progress.
Strategic Agent	Evaluates vision, long-term market direction, growth potential, and strategic positioning. Frames the big picture.
Business Agent	Analyses market size, competitive dynamics, business model viability, monetisation strategy, and revenue opportunities.
Financial Agent	Reviews cost structure, revenue projections, ROI calculations, break-even analysis, and funding requirements.
Technical Agent	Assesses technical feasibility, proposed architecture, technology stack choices, scalability, and implementation risk.
Customer Agent	Maps target user personas, evaluates customer adoption likelihood, addresses UX considerations and customer concerns.
Risk Agent	Identifies and scores legal, regulatory, operational, financial, and reputational risks with severity classifications.
Data Agent	Analyses uploaded datasets — profiling, trend detection, anomaly detection, statistical insights, and forecasting.
Devil's Advocate	Challenges all assumptions, generates counter-arguments, simulates failure scenarios, and surfaces hidden weaknesses.
Reviewer Agent	Collects all reports, detects contradictions, resolves conflicts, scores confidence per finding, and generates the executive recommendation.

08 Dynamic Agent Selection

The CEO Agent analyses each submission and activates only the agents relevant to that specific input type. Unused agents are never invoked. This reduces token cost, latency, and noise in the final output — while increasing relevance and depth.

INPUT TYPE	AGENTS ACTIVATED
Startup Idea / Business Proposal	Strategic · Business · Financial · Customer · Risk · Devil's Advocate
Dataset Analysis	Data · Financial · Business · Risk
Research Paper	Technical · Strategic · Risk · Devil's Advocate
Product Proposal	Business · Technical · Customer · Risk · Strategic
Financial Report	Financial · Risk · Business · Strategic
Marketing Strategy	Business · Customer · Strategic · Risk · Devil's Advocate
Technical Architecture Review	Technical · Risk · Strategic · Devil's Advocate
Investment / Due Diligence	Financial · Business · Strategic · Risk · Customer
Organizational Decision	Strategic · Risk · Business · Customer · Devil's Advocate

★ Reviewer Agent + CEO Summary Agent are always activated, regardless of input type.

09 Collaboration Model

Agents do not operate independently forever. The boardroom discussion follows a structured three-round protocol that mirrors how real executive panels reach consensus.

ROUND	PHASE	DESCRIPTION
Round 1	Independent Analysis	Each selected agent receives the user input via the shared RAG workspace and produces a fully independent expert report. No agent sees any other agent's output at this stage.
Round 2	Peer Review	Every agent reads the other agents' Round 1 reports. Each agent identifies agreements, flags contradictions, and updates its own analysis with new context discovered in peer findings.
Round 3	Challenge & Refine	Agents may challenge specific claims from other agents, reinforce consensus positions, surface new concerns raised by the collective analysis, and submit final refined positions.
Final	Synthesis & Verdict	The Reviewer Agent collects all Round 3 outputs, detects remaining conflicts, builds consensus, assigns per-claim confidence scores, and generates the final recommendation with reasoning.

Decision Outcomes

GO	High confidence · All major agents aligned · Risks manageable · Clear opportunity
GO WITH MODIFICATIONS	Moderate confidence · Specific concerns raised · Conditional approval with action items
NEEDS FURTHER RESEARCH	Low confidence · Critical unknowns · Missing data · Recommend additional analysis
NOT RECOMMENDED	Significant risks · Fundamental flaws identified · Multiple agents opposed

Each outcome includes: confidence score (0–100%), full reasoning chain, supporting evidence from agents, and a prioritised action plan.

10 Output & Report Types

REPORT NAME	PRODUCED BY	CONTENTS
Strategic Report	Strategic Agent	Vision, long-term potential, market positioning, strategic opportunities
Business Report	Business Agent	Market sizing, competitor analysis, business model review, revenue strategy
Financial Report	Financial Agent	Cost analysis, ROI, revenue forecast, profitability, budget requirements
Technical Report	Technical Agent	Feasibility, architecture, technology recommendations, scalability review
Customer Report	Customer Agent	User personas, adoption analysis, UX recommendations, customer concerns
Risk Report	Risk Agent	Legal, operational, financial, and reputational risks with severity scores
Data Report	Data Agent	Dataset profiling, trend analysis, statistical insights, forecasting
Contrarian Report	Devil's Advocate	Assumptions challenged, failure scenarios, counter-arguments, weak points
Boardroom Discussion Report	All Agents	Full multi-round discussion transcript with debates, agreements, conflicts
Executive Summary	Reviewer Agent	Consolidated recommendation, confidence score, key findings, action plan

Export Formats

All reports are available for export in three formats, styled as professional management consulting deliverables — not generic AI output.

PDF	DOCX	PPT
Formatted via ReportLab. Board-meeting ready. Printable & shareable.	Microsoft Word compatible. Editable by stakeholders. Track changes enabled.	Slide deck via python-pptx. Presentation ready. Investor pitch format.

11 System Architecture

AI Boardroom is built on a 14-layer architecture designed for enterprise-grade reliability, horizontal scalability, and full observability. Each layer is independently deployable.

LAYER	TECHNOLOGY	RESPONSIBILITIES
1 — Presentation	Next.js · React · TypeScript · R3F	Dashboard, Boardroom View, Reports, Analytics, Real-time WebSocket updates, 3D agent visualization
2 — API Layer	FastAPI · Python · Uvicorn · WebSocket	Authentication, file uploads, agent requests, session management, report generation, streaming
3 — CEO Orchestrator	LangGraph · Shared State	Input classification, agent selection, workflow management, progress tracking, workspace creation
4 — Agent Layer	9 Specialized AI Agents	Parallel independent analysis, peer review, debate rounds, consensus building
5 — AI Model Layer	OpenAI · Anthropic · Gemini	Multi-provider routing, cost optimization, load balancing, automatic fallback on provider failure
6 — Knowledge Layer	Qdrant · RAG Pipeline	Document embeddings, semantic search, session memory, agent memory, shared boardroom context
7 — Database Layer	PostgreSQL · SQLAlchemy	Users, projects, sessions, agent outputs, discussion logs, reports, confidence scores, audit trails
8 — File Storage	Amazon S3 / S3-compatible	PDFs, DOCX, PPT, Excel, JSON uploads; generated report assets; signed URL access control
9 — Collaboration	Agent Discussion Engine	Round 1 opinions, Round 2 peer review, Round 3 debate, consensus building, history storage
10 — Report Generation	ReportLab · python-pptx	10 report types generated per session; PDF, DOCX, PPT; board-grade visual formatting
11 — Auth & Security	OAuth 2.0 · JWT · RBAC	Google + Microsoft login, role-based access control, route guards, data encryption, API security
12 — Monitoring	PostHog · Agent Logs	Prompt logs, decision tracking, audit trails, error monitoring, performance metrics, agent KPIs
13 — Analytics	PostHog · Custom Dashboards	User behaviour, agent performance, report quality metrics, usage statistics, business metrics
14 — Deployment	Docker · Kubernetes · CI/CD	Auto-scaling, load balancing, blue-green deployments, disaster recovery, 99.9% uptime SLA

12 Database Architecture

PostgreSQL — Relational Store	Qdrant — Vector Store	S3 — Object Store
• users • projects • sessions • agent_outputs • discussion_logs • reports • confidence_scores • audit_trails	• document_embeddings • knowledge_chunks • session_memory • agent_memory • semantic_search_index	• uploaded PDFs / DOCX / PPT • CSV & Excel datasets • generated report files • project media assets • signed URL access control

Key PostgreSQL Tables

TABLE	KEY COLUMNS
users	id, email, name, role, oauth_provider, created_at, last_login
projects	id, user_id, title, description, input_type, status, created_at
sessions	id, project_id, agents_selected, round_count, status, started_at
agent_outputs	id, session_id, agent_name, round_number, report_content, tokens_used
discussion_logs	id, session_id, round, agent_from, agent_to, message, timestamp
reports	id, session_id, report_type, file_url, generated_at, format
confidence_scores	id, session_id, claim, score, supporting_agents, conflicting_agents
audit_trails	id, user_id, action, resource, timestamp, ip_address, metadata

13 AI Model Layer

AI Boardroom uses a multi-provider AI gateway with intelligent model routing. No single AI provider is a point of failure. Cost, capability, and latency are balanced dynamically per agent and per task type.

MODEL	ROLE	USE CASE
OpenAI GPT-4o	Primary	General analysis, business, strategic, customer agents
OpenAI GPT-4o-mini	Economy	Lower-cost routing for simpler sub-tasks and summaries
Anthropic Claude	Primary	Long-context documents, nuanced reasoning, risk analysis
Google Gemini Pro	Fallback	Backup provider when OpenAI or Anthropic are unavailable
Open Source (future)	Optional	Self-hosted models for data-sensitive enterprise deployments

FEATURE	IMPLEMENTATION
Model Routing	CEO Agent selects the optimal model per agent based on task type, context length, and cost budget
Cost Optimization	GPT-4o-mini used for summarization and simple tasks; GPT-4o / Claude for deep analysis
Fallback Mechanism	Automatic failover to secondary provider if primary returns error or rate-limit
Load Balancing	Requests distributed across provider API keys to avoid rate-limit throttling
Context Management	Shared boardroom memory ensures agents only retrieve the context they need via RAG
Streaming	All agent responses stream to the frontend via WebSocket for live boardroom feel

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Performance & Scalability

OPTIMIZATION	IMPLEMENTATION	BENEFIT
Dynamic Agent Selection	Only relevant agents activated per session	Up to 60% reduction in token cost & latency
Parallel Execution	All agents run simultaneously via async task queues	Analysis time equivalent to slowest agent, not sum
Shared RAG Memory	Documents embedded once; all agents retrieve via Qdrant	Zero duplicate document processing
WebSocket Streaming	Real-time agent status and output pushed to UI	Live boardroom experience with no polling
Caching Layer	Session-level response caching for repeated queries	Sub-200ms response for cached endpoints
Horizontal Scaling	Kubernetes auto-scaling agent pods independently	Scale agent workers without scaling full stack
Model Fallback	Multi-provider routing with automatic failover	Zero AI single-point-of-failure
Async Task Queue	Celery + Redis for background agent execution	Non-blocking API; UI updates via WebSocket events

Scalability Targets



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Technology Stack

AI Boardroom is built on a carefully selected modern stack — optimised for developer velocity, enterprise reliability, and the specific demands of multi-agent AI systems.

TOOL / LIBRARY	CATEGORY	PURPOSE IN PLATFORM
■ FRONTEND		
Next.js 14	Frontend Framework	App router, server components, SSR/SSG for fast, SEO-ready pages
React 18	UI Library	Component-based UI with concurrent rendering and Suspense
TypeScript	Language	Type safety across entire frontend codebase; fewer runtime errors

Tailwind CSS	Styling	Utility-first CSS; dark/light mode via CSS variables
shadcn/ui	Component Library	Accessible, unstyled Radix UI primitives — fully customizable
Framer Motion	Animations	Spring-based motion for agent transitions, panel reveals, state changes
React Three Fiber	3D Rendering	Declarative Three.js in React — powers the live 3D boardroom view
Three.js + Drei	3D Utilities	Lighting, camera controls, geometry helpers for boardroom scene
Zustand	State Management	Lightweight global state for session, agent status, and UI state
WebSockets (native)	Real-Time	Persistent connection for live agent streaming and progress updates
■ BACKEND		
FastAPI	API Framework	High-performance async Python REST API; auto OpenAPI docs
Python 3.12	Language	Agent logic, orchestration, report generation, data processing
Uvicorn	ASGI Server	Lightning-fast ASGI server; handles WebSocket + HTTP concurrently
LangGraph	Agent Orchestration	Stateful multi-agent graph — manages rounds, shared state, routing
Celery	Task Queue	Async background execution of agent analysis tasks
Redis	Cache + Broker	Celery message broker; session cache; real-time pub/sub for events
SQLAlchemy	ORM	Database models, migrations, query abstraction for PostgreSQL
Pydantic v2	Validation	Request/response schema validation; strongly typed agent outputs
python-multipart	File Handling	Multipart file upload parsing for PDF, DOCX, CSV, Excel ingestion
ReportLab	PDF Generation	Board-grade PDF report creation with custom layout and branding
python-pptx	PPT Generation	PowerPoint export of all 10 report types for presentation use
python-docx	DOCX Generation	Microsoft Word export; editable by stakeholders post-generation
■ AI PROVIDERS & MEMORY		
OpenAI GPT-4o	Primary AI Model	Deep analysis agents — Strategic, Business, Financial, Customer
OpenAI GPT-4o-mini	Economy Model	Cost-optimised routing for summarization and lightweight sub-tasks
Anthropic Claude	Primary AI Model	Long-context documents, nuanced risk analysis, Devil's Advocate
Google Gemini Pro	Fallback Model	Automatic failover when OpenAI or Anthropic are rate-limited
OpenAI Embeddings	Embedding Model	text-embedding-3-large for document chunking and Qdrant storage
Qdrant	Vector Database	High-performance vector search; stores all document embeddings
LangChain	RAG Utilities	Document loaders, text splitters, retrieval chain helpers

■ DATABASES & STORAGE

PostgreSQL 16	Relational DB	Primary structured store: users, sessions, reports, audit trails
Amazon S3 / MinIO	Object Storage	File uploads, generated report assets; signed URL access control
Redis 7	In-Memory Store	Caching, Celery broker, WebSocket pub/sub, session tokens
Qdrant	Vector Store	Document embeddings, agent memory, semantic retrieval per session

■ AUTH · INFRASTRUCTURE · MONITORING

OAuth 2.0 / JWT	Authentication	Google + Microsoft social login; JWT access + refresh token flow
Docker	Containerisation	All services containerised; consistent dev/staging/prod environments
Kubernetes + Helm	Orchestration	Auto-scaling pods per agent type; Helm charts for deployment
GitHub Actions	CI/CD	Automated test, build, and deploy pipeline on every PR merge
NGINX Ingress	Load Balancing	TLS termination, routing, rate limiting at the cluster edge
Prometheus + Grafana	Observability	Metrics collection and real-time dashboards for system health
Sentry	Error Tracking	Frontend and backend exception capture with full stack traces
PostHog	Product Analytics	User behaviour, feature adoption, agent performance analytics

16 Visual Design System

The platform aesthetic sits at the intersection of Apple Vision Pro spatial computing, a modern executive boardroom, and a cinematic AI workspace. Every design decision reinforces authority, trust, innovation, and premium intelligence. Users should feel they are presenting to a panel of real experts — not uploading a file to another AI tool.

Colour Palette

Deep Navy #060E1E	Card Surface #0F1C2E	Elec. Blue #2563EB	Elec. Cyan #00D4FF	Prem. Gold #FFD700	Risk Red #FF4D4D

Dark Mode & Light Mode

MODE	SPECIFICATION
Dark Mode (Primary)	#020617 BG · #0F172A Cards · #1E293B Borders · #00D4FF Highlights
Light Mode	#F8FAFC BG · #FFFFFF Cards · #CBD5E1 Borders · #2563EB Accents
Auto Detection	Respects OS preference; user can override in Settings

3D & Motion Experience

ELEMENT	SPECIFICATION
3D Boardroom View	React Three Fiber + Three.js virtual boardroom table; each seat represents one active agent
Agent State Animation	Active agent seats glow with their accent color; pulsing animation during analysis; idle when complete
Floating Glass Panels	Glassmorphism cards with depth effect via CSS backdrop-filter and Three.js depth layers
Cinematic Lighting	Directional and ambient lighting in 3D scenes; dramatic shadows for premium feel
Framer Motion	All UI transitions use spring-based Framer Motion animations — purposeful, never decorative-only
Streaming Indicators	Live token-by-token streaming for agent reports; typing indicators during analysis
Report Style	Reports styled as McKinsey/BCG-grade management consulting deliverables — not chat output

17 Security & Compliance

DOMAIN	REQUIREMENT
Authentication	OAuth 2.0 with Google and Microsoft; email/password with bcrypt hashing; session JWT tokens
Authorization	Role-Based Access Control (RBAC); Admin and User roles; route-level guards on all protected endpoints
Data Encryption	AES-256 at rest for all stored documents, reports, and embeddings; TLS 1.3 for all data in transit
Audit Trails	Immutable logs of all agent prompts, model responses, decision outputs, and user actions with timestamps
API Security	Rate limiting per user/IP; API key rotation; CORS policy enforcement; all inputs sanitised and validated
File Security	Signed S3 URLs with expiry; no public bucket access; per-user scoped storage; virus scanning on upload
Session Security	JWT with short expiry + refresh tokens; session invalidation on logout; concurrent session limits
Password Recovery	Secure email-based OTP flow; tokens expire in 15 minutes; cannot reuse last 5 passwords

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Non-Functional Requirements

CATEGORY	REQUIREMENT	TARGET / SPECIFICATION
Performance	API Response Time	< 200 ms per endpoint under normal load
Performance	First Agent Output	< 30 seconds for standard text submissions
Performance	Full Boardroom Analysis	< 3 minutes for mixed document + text sessions
Performance	Dashboard Initial Render	< 3 seconds; Lighthouse score > 90
Performance	Report Generation	< 60 seconds per full session (all 10 reports)
Reliability	Uptime SLA	99.9% with automated failover and health checks
Reliability	AI Model Fallback	Automatic provider switch on failure — zero service interruption
Reliability	Data Refresh	WebSocket streaming; agent progress updated every 2 seconds
Reliability	Graceful Degradation	If one agent fails, session continues; failure noted in report
Scalability	Concurrent Boardroom Sessions	10,000+ simultaneous sessions in Phase 3
Scalability	Max File Upload	50 MB per file; 200 MB per session
Scalability	Agents Per Session	Up to 9 agents in parallel; Kubernetes pod scaling per agent
Scalability	Document Context	Up to 500,000 tokens per boardroom session via RAG chunking
Security	Auth Standard	OAuth 2.0 + JWT; MFA optional in v1.0; required in v2.0
Observability	Logging Coverage	100% of agent prompts, responses, decisions — stored with ms precision
Portability	Deployment Target	Docker + Kubernetes; cloud-agnostic (AWS / GCP / Azure)

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Project Structure

FRONTEND	BACKEND	INFRASTRUCTURE & DB
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frontend/ ■■■ app/ ■ ■■■ (auth)/ ■ ■■■ dashboard/ ■ ■■■ projects/ ■ ■■■ boardroom/ ■ ■■■ reports/ ■ ■■■ settings/ ■■■ components/ ■ ■■■ boardroom-3d/ ■ ■■■ agents/ ■ ■■■ reports/ ■ ■■■ shared/ ■■■ lib/ ■ ■■■ api/ ■ ■■■ store/ ■■■ public/	backend/ ■■■ api/ ■ ■■■ routes/ ■ ■■■ middleware/ ■■■ agents/ ■ ■■■ ceo/ ■ ■■■ strategic/ ■ ■■■ business/ ■ ■■■ financial/ ■ ■■■ technical/ ■ ■■■ customer/ ■ ■■■ risk/ ■ ■■■ data/ ■ ■■■ devils_advocate/ ■ ■■■ reviewer/ ■■■ orchestrator/ ■■■ memory/ ■■■ rag/ ■■■ reports/ ■■■ storage/ ■■■ database/ ■■■ monitoring/	infrastructure/ ■■■ docker/ ■ ■■■ frontend/ ■ ■■■ backend/ ■■■ kubernetes/ ■ ■■■ deployments/ ■ ■■■ services/ ■ ■■■ ingress/ ■■■ ci-cd/ ■ ■■■ github-actions/ ■■■ deployment/ database/ ■■■ postgres/ ■ ■■■ migrations/ ■■■ qdrant/ ■■■ collections/
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Out of Scope — v1.0

The following features are explicitly excluded from the v1.0 release and are planned for v2.0 or later. This boundary is set to maintain delivery focus and architectural integrity.

OUT-OF-SCOPE ITEM	TARGET VERSION	NOTES
Native Mobile App (iOS / Android)	v2.0	Progressive web app covers mobile in v1.0
Team / Multi-User Collaborative Workspaces	v2.0	Shared sessions with multiple human collaborators
Real-Time Voice Synthesis / Audio Output	v2.0	Agent reports read aloud; voice interaction
Physical System Integration	v3.0	Automated actuation based on AI recommendations
Custom Model Fine-Tuning	v3.0	Domain-specific fine-tuned agent models
Third-Party ERP / CRM Integration	v2.0	Salesforce, SAP, HubSpot direct connectors
Multi-Language Interface	v2.0	English-only for v1.0 launch
White-Label / Multi-Tenant SaaS	v2.0	Branded deployments for enterprise clients
Deep Learning Forecasting (LSTM etc.)	v3.0	Custom neural time-series models for Data Agent
MFA / SSO Enforcement	v2.0	Optional in v1.0; mandatory admin feature in v2.0

The Final Vision

AI Boardroom is not another AI wrapper. It is a fundamentally new category of software — a virtual decision intelligence layer that sits between raw information and confident action. When a user submits their idea, they should feel the weight of an entire boardroom's attention: strategic minds, financial rigor, technical scrutiny, legal caution, and adversarial challenge — all converging into a single, transparent, accountable recommendation.

The platform democratizes access to board-level intelligence — making what was once available only to well-funded enterprises (panels of expert consultants, strategy sessions, financial reviews) accessible to any founder, analyst, researcher, or organization in minutes.

"Build an enterprise-grade AI decision intelligence platform that delivers trustworthy, explainable, and actionable decisions — through a single immersive product called AI Boardroom."



AI Boardroom · Product Requirements Document · v1.0 · 2025